EE5203 L09 Worksheet

ADC and sensor acquisition - Basri Erdogan

Expected time: 30-45 minutes **Policy:** Attempt every question before consulting the worked solutions.

12-bit ADC, 3.3 V reference, `raw` in 0…4095 throughout.

Part A - Counts, volts, and resolution

1. Write the counts→millivolts formula, and state the LSB size in millivolts.
2. Complete the table using integer arithmetic only.

`raw`	mV	LED duty (`×999/4095`)	Servo µs (`1000 + ×1000/4095`)
0			
1024			
2867			
4095			

3. `mV = (raw / 4095) * 3300;` — explain exactly what this computes for `raw = 2000`, and give the corrected expression.
4. Would 16-bit resolution on the same 3.3 V reference help you distinguish two voltages 0.5 mV apart? Compute the LSB for both 12-bit and 16-bit and justify your answer — including one reason the extra bits might not help in practice.
5. A team wants to measure a 0–5 V sensor with this ADC. Explain what happens if they connect it directly, and describe a resistor divider that brings 5 V to just under 3.3 V. Give the two resistor values and the resulting `raw` for a 5 V input.

Part B - Configuration

6. Give the `MODER` bit pattern for an analog pin, and say what else changes inside the pin when analog mode is selected.
7. In one sentence each: what do **sampling time** and **conversion time** mean, and which one do you choose in CubeMX?
8. Explain why a high-impedance source needs a longer sampling time. What symptom appears if it is too short *and you are switching channels*?
9. There is one ADC and sixteen channels. Describe what actually happens when `adc_read(ADC_CHANNEL_0)` is followed by `adc_read(ADC_CHANNEL_1)`, and state one consequence for a project that must correlate two sensors.
10. Contrast **single** and **continuous** conversion mode. Give one situation where each is the right choice.

Part C - Reading and flags

11. What sets `EOC`, and what clears it? Which HAL call does the clearing?
12. A polling loop reads `ADC1->DR` twice per conversion “to be safe”. Explain what the second read returns and why the loop then drifts out of step.
13. Why is `HAL_ADC_PollForConversion` unacceptable inside an interrupt callback? Give the alternative and name the flag it relies on.
14. List the three flag families you have met so far, their peripherals, and what clears each one.

Part D - From counts to control

15. Write the loop body that reads the pot every 10 ms from the L07 tick and sets the L08 LED duty. No `HAL_Delay()`.
16. `duty = (raw * 999u) / 4095u;` — show that this cannot overflow a `uint32_t`, and state what would happen with `uint16_t` intermediates.
17. Write `servo_from_raw(uint16_t raw)` returning a pulse width clamped to the 1000–2000 µs contract. Give the value for `raw = 2867`.
18. The knob is noisy: the last count flickers between two values and the servo buzzes. Give **two** different mitigations, one in software and one in hardware, and state a cost of each.

Part E - Sensors and evidence

19. An LDR in series with a 10 kΩ resistor between 3V3 and GND, junction to PA1. As the room gets **darker**, does `raw` rise or fall? Justify from the divider, given that an LDR's resistance rises in darkness.
20. Your night-light toggles LD2 rapidly when the light sits near the threshold. Name the phenomenon and give a fix that does not change the sensor. Then say what you would set the two thresholds to if the trip point is `raw = 2000`.
21. Fill in the evidence you would show for each claim.

Claim	SFR observation
The pin is in analog mode	
A conversion has completed	
Reading the result clears the flag	
The converter tracks the knob	
Continuous mode is genuinely free-running	

22. Describe an SFR-view experiment that demonstrates continuous mode without changing a line of code, and state what you would see that you do not see in single mode.

Exit self-assessment

- ☐ I can convert counts to millivolts in integer arithmetic.
- ☐ I can justify a sampling time from source impedance.
- ☐ I can explain what clears `EOC`.
- ☐ I can map a reading to a duty or a servo pulse, clamped.
- ☐ I can diagnose threshold chatter and fix it.